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Characterization of ARDESIA: a 4-channel SDD X-ray spectrometer for synchrotron measurements at high count rates

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Abstract: ARDESIA is an SDD-based, X-ray spectrometer, optimized for synchrotron measurements that require high count rates (> 1 Mcounts/s/channel) and good energy resolution (< 130 eV of FHWM Mn-K$_\alpha$ line at optimum shaping time, ≤ 200 eV at short shaping times). The main target applications are XRF and XAFS techniques. The detection module consists of 2 × 2 pixel monolithic SDD (5 mm pitch) coupled with a 4-channel version of the CUBE CMOS preamplifier. The mechanical structure of the instrument has been realized to fit inside a sample chamber with a finger-like structure. The system grants proper cooling (-40°C) and operation in vacuum. ARDESIA is also equipped with two auxiliary electronic boards: one for SDD powering and biasing and the other for closed-loop driving of two Peltier TECs. The output signals of the instrument are processed by digital pulse processors using short pulse processing times in order to achieve good spectroscopic performance at high count rates. Two different measurement campaigns were

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performed at synchrotron beamlines to assess the performance of the instrument. At the LNF DAΦNE-Light DXR1 soft X-ray beamline, XRF measurements on low atomic number elements (down to C-K line, 277 eV) demonstrated the good energy resolution of the spectrometer and made possible to acquire the first XAFS spectrum of Silicon K-edge in a Pyrex^{TM} glass sample. At the LISA CRG beamline at ESRF — Grenoble, XAFS measurements on different samples, such as kesterite and protochabournéite, were performed, demonstrating the high count rate capability and stability of the instrument over time.

KEYWORDS: Instrumentation for synchrotron radiation accelerators; X-ray fluorescence (XRF) systems; Solid state detectors

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1 Introduction

X-ray synchrotron radiation beamlines provide very powerful tools to perform atomic and electronic structural investigations. Powerful tools require also the development of new instrumentation [1], like detectors for X-rays detection [2–7]. Within our collaboration, the DAΦNE-Light DXR1 soft X-ray beamline at Frascati National Laboratory (LNF, Rome) is characterized by a high-flux beam of soft X-rays [8], whose energy ranges from 0.9 keV to 3 keV. This beamline is normally used for X-ray Absorption Fine Structure (XAFS) measurements on samples containing low-Z atomic elements, benefitting from the high energy resolution of well-aligned, energy-controlled X-ray beams at the output of a double crystal monochromator. XAFS can be performed in transmission mode, placing the sample between two ionization chambers to measure the intensity of the incoming and the transmitted beams using intensity detector [9, 10]. This technique is relatively straightforward and can provide excellent counting statistics. However, the transmission mode technique is restricted to samples with specific characteristics: uniformity, thickness of $\sim$ 2 times the absorption length, and absence of pinholes. Measurements in transmission mode can be used for samples with a concentration of the element of interest larger than 10% in weight [11]. If supported or very diluted samples have to be tested, XAFS measurements are usually performed in fluorescence mode. In fluorescence mode, a wide solid angle of the detector is essential, in order to collect most of the fluorescence isotropic emissions produced in the region where the scattering is minimum, improving the counting statistics of the spectrum [12]. Energy resolution is also very important because it allows suppressing the spectrum background given by the scattered X-rays or fluorescence lines of other elements. Finally, a high-count rate (few Mcounts/s) capability gives the possibility to perform the measurement with better counting statistics or shorter acquisition times, fully exploiting the high flux of synchrotron radiation.

Based on these requirements for fluorescence detectors, the ARDESIA X-ray detection system has been developed, focused on XRF and XAFS applications in fluorescence mode using synchrotron radiation. Currently, the 4-channel instrument prototype has been developed and tested

in two different synchrotron facilities: the LNF DAΦNE-Light DXR1 soft X-ray beamline, already previously mentioned, and the LISA BM-08 beamline at ESRF, where larger X-ray energies are available. In this paper, the design and the building blocks of the 4-channel instrument will be presented, together with the results of extensive experimental tests and validation measurements carried out at LNF and ESRF, showing its full functionality and performance.

2 The ARDESIA spectrometer

The mechanical structure of the instrument has been realized to properly fit inside a sample chamber with a finger-like structure, as shown in figures 1 and 2. The outer diameter of the vacuum-tight probe is 38 mm, while the internal metal finger has a diameter of 20 mm. The backbone of the structure consists of three parts: (a) the detection head, (b) the cold finger, and (c) the base with connectors and a water-cooled heatsink. The interchangeable detection module is screwed on the tip of the cold finger. These three parts are cooled using a system based on two Peltier thermoelectric coolers (TEC), a $15 \times 15\,\mathrm{mm}^2$ double-stage Peltier cell placed in between the detection module and the cold finger and a $55 \times 55\,\mathrm{mm}^2$ single-stage Peltier cell placed in between the cold finger and the base heatsink. Using this strategy, when the base heatsink is cooled to 5°C by the chiller, the instrument can reach a temperature as low as −40°C at the detection module. The external single-structured stainless steel covering tube is attached to the heatsink to protect all the internal parts and maintain static vacuum condition (10^{-6}–10^{-2} mbar depending on the pump). Static vacuum condition inside the instrument is crucial, not only to prevent ice formation on the surface of the detector during cooling but also to maintain the vacuum environment in the sample chamber where the instrument is placed. The entrance window, made by AP5 ultra-thin polymer, Be, or Kapton and graphite sheet, is placed at the tip of the covering tube. The critical output signals from the detection module are buffered by the internal electronics to preserve the signal integrity, before then delivered to the outside of the instrument and processed by commercial digital pulse processors. Two vacuum-grade Fischer®connectors, providing the interface for power supplies and temperature monitoring, are attached at the base of the heatsink. Two additional external electronic units manage power supply and cooling, making the system a stand-alone and easily transportable instrument. Besides combining low noise with high count rates (fast peaking times), the main challenges in the design of the instrument were: (*i*) compactness, (*ii*) signal integrity, preserving the ultra-low noise performance (especially critical at short shaping times) also in the noise environment of a synchrotron beamline. The details about the internal and external electronics are illustrated in the following sections.

2.1 Detection module

The 4-channel ARDESIA detection module features a total area of $16 \times 16\,\mathrm{mm}^2$. The detectors is a 2×2 SDD monolithic array (figure 3), having a pixel pitch of 5 mm and a thickness of 450 μm. SDD is a suitable choice for high-count rate applications, thanks to the full sideward depletion and small anode capacitance, independent of the chosen active area [13]. The detectors are produced by Fondazione Bruno Kessler (FBK, Trento, Italy) with a low-leakage technology process ($< 200\,\mathrm{pA/cm}^2$ at room temperature, RT) [14], which allows achieving excellent energy resolution. Two different pixel geometries are produced: the circular pixels ($20\,\mathrm{mm}^2$ per pixel,

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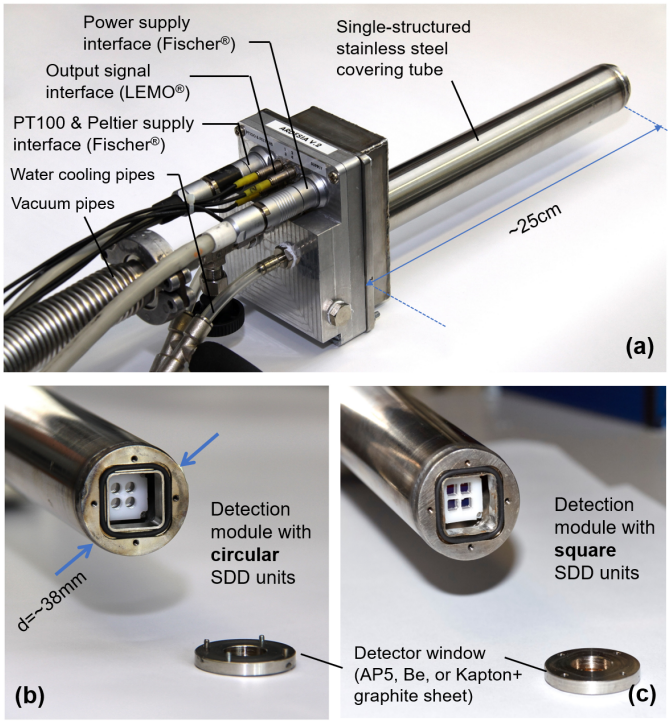


Figure 1. Photograph of the assembled ARDESIA X-ray spectrometer: external view (a) from the rear and from the head (without the entrance window) showing circular (b) and square (c) pixels.

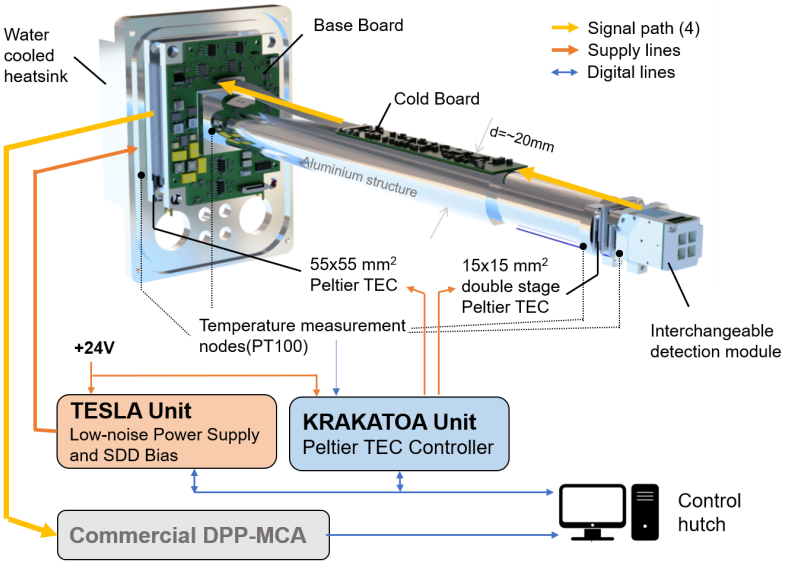


Figure 2. Architecture of the ARDESIA spectrometer.

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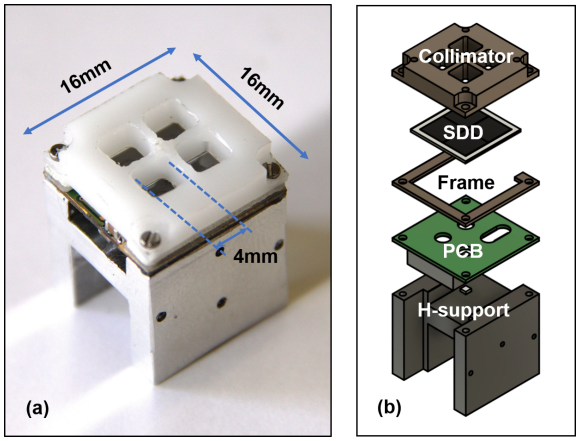


Figure 3. Interchangeable 4-channel detection module: (a) zoom and (b) exploded-view.

collimated to 12.56 mm^2) and the square pixels (25 mm^2 per pixel, collimated to 16 mm^2). These choices offer different tradeoffs between fill ratio and uniformity of the charge drift field. The detector with squared elements has wider active area but higher maximum drift time, due to the lower intensity of the electric field at the corners of the diagonals of the SDD pixels. The detector chip is glued directly to the PCB, guided by the plastic frame at the border.

The detector current outputs are read out by the CUBE ASIC, a charge integrating preamplifier [15]. The CUBE is a low-noise CMOS-based preamplifier that is meant to be used with detectors without integrated JFET [16]. For each detection module, a 4-channel preamplifier is placed close to the detector, on the other side of the PCB and the anodes of the detector are connected by direct wire bonding to the gates of the preamplifier, through dedicated holes on the PCB. The input MOSFET of CUBE preamplifier is characterized by high transconductance, thus compensating for parasitic capacitance associated with wire bonding and reducing the series noise contribution of the electronic noise. For this reason, the SDD-CUBE solution is particularly suited for high count rate applications [17]. The CUBE preamplifier generates ramp signals with step pulses corresponding to the event generation, which are reset by external signals. When the SDD and the CUBE are biased properly, the CUBE generates step pulses with the average rise time of $\sim$ 40 ns and a conversion gain of $\sim$ 1.7 mV/keV.

The collimator is placed over the detector, so that no charge is generated in the pixel boundary regions, avoiding charge-sharing issues. A Delrin^{TM} plastic collimator is used for energies up to 6 keV, while a molybdenum collimator is used for energies up to 20 keV. The PCB, where the detector and preamplifier chips are attached, and the collimator are screwed to an aluminum H-support that acts as heatsink. The small size of the detection module makes it possible to flexibly arrange multiple modules together to cover larger solid angles without increasing dead areas too much.

The characterization of the detection module has been performed and reported at both low and high input count rates (ICR) using an ^{55}Fe laboratory source [18]. However, new results from improved detection module are reported in this paper. Low ICR characterization is required to obtain the best energy resolution at optimum shaping time given by the detection module, eliminating other

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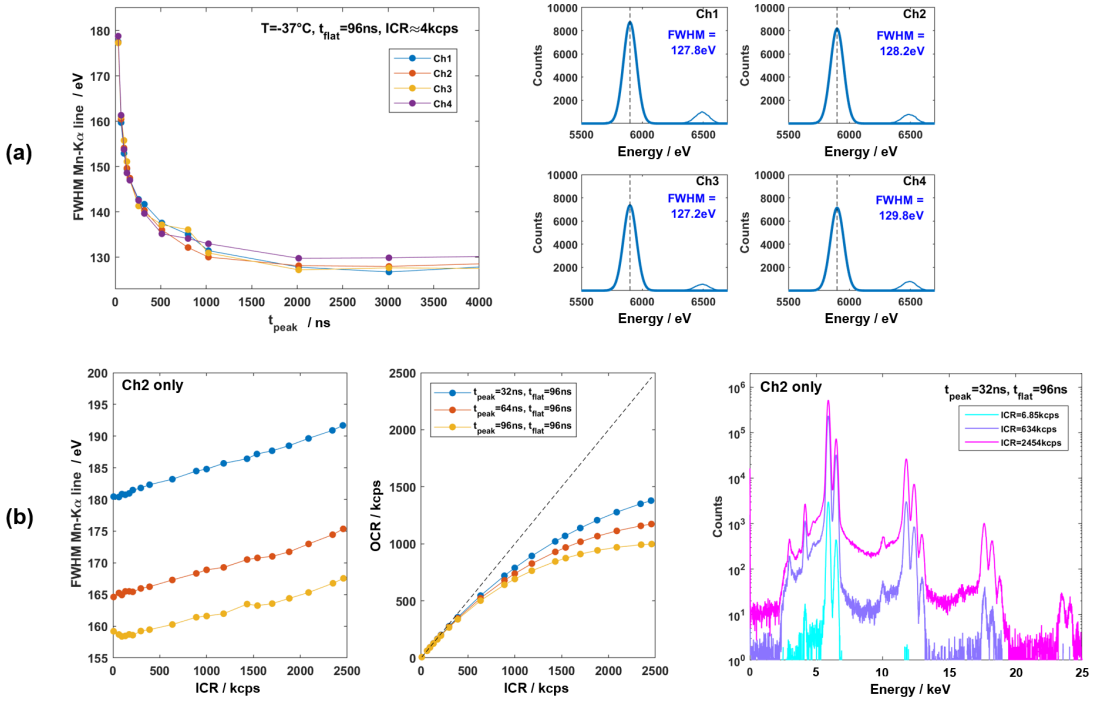


Figure 4. Detection module characterization: energy resolution at low ICR (a), and resolution and throughput at high ICR (b).

factors like pile-up conditions. An ^{55}Fe laboratory source was placed within a certain distance from the detector window to obtain an ICR of 4 kcounts/s per channel. For the DPP (Dante by XGLab, Italy) trapezoidal filter settings, t_{flat} = 96 ns was used to suppress almost completely the ballistic deficit effect due to the finite rise time of the step pulses at the output of the system SDD-CUBE. In figure 4a (left), the resolution of 179 eV (Mn-K$_\alpha$ line FWHM), 15.8 e- ENC was obtained at the shortest peaking time (t_{peak} = 32 ns, t_{flat} = 96 ns), more than 20eV better than what has been reported in [18], while the resolution of 127.2 eV, 5.3 e- ENC was obtained at optimum t_{peak} = 3 µs using the best performing channel. The spectra obtained at the optimum peaking time are shown in figure 4a (right).

The source was then placed closer to the center point of one of the channel to increase the total ICR up to 2.5 Mcounts/s. In figure 4b, the FWHM and OCR are plotted as a function of the ICR per channel. At t_{peak} = 32 ns, the resolution increased slightly to 192 eV, corresponding to 17.7 e- rms, at ICR = 2.5 Mcounts/s. A maximum OCR of 1.37 Mcounts/s was obtained at ICR = 2.5 Mcounts/s. Therefore, it is expected that a total OCR of about 5.5 Mcounts/s is achievable with the 4-channel ARDESIA instrument using this DPP. By increasing t_{peak} to 96 ns, the OCR reduces to 1 Mcounts/s per channel but with an improvement of the energy resolution to 167.6 eV FWHM.

2.2 Internal and auxiliary electronics

The signals coming out from the detection module need to be buffered to maintain their integrity and provide immunity against possible external disturbances in the beamline, such as stray currents

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from un-optimized grounding connection, electromagnetic interference (EMI), or vibration-induced disturbance. For this purpose, the internal electronics has two low-noise signal buffers, one unity-gain buffer thermally attached on the cold finger (on the Cold Board in figure 2), close enough to the detection module to minimize the load capacitance from the wire parasitics, followed by another buffer with a gain of 3 (on the Base Board in figure 2). The gain of the second buffer was chosen to optimize the signal to noise ratio and to preserve the rise time of the step pulses of the events. The buffered output signals reach the DPP through vacuum-grade LEMO connectors. Besides the signal buffers, low-noise voltage regulators for SDD-CUBE biasing and CUBE reset pulse generation are also placed inside the instrument. The grounding scheme has been considered as a key aspect of the ARDESIA electronics design. Two important principles were used in designing the grounding schemes. First, the output signal, from the source (detection module) to the destination (electrical interfaces and DPPs), should maintain the same grounding reference. Second, all separated ground planes must be finally connected together in a star topology to avoid possible ground loops.

As a stand-alone instrument, ARDESIA is also equipped with two auxiliary electronic units: TESLA, for powering the internal electronics, and KRAKATOA, for temperature monitoring and controlling Peltier TEC to achieve the desired low temperature for measurements. TESLA provides a ±12 V clean supply for internal electronics and high voltages for the SDD biasing (allowing also to program the desired detector power-on sequence). TESLA was designed adopting power supply filtering techniques to avoid bypassing the disturbances from the outside to the critical signal paths. KRAKATOA provides an automatic cooling process for the instrument, based on the PT100 temperature readings on 4 different spots: detection module, cold fingertip, cold finger base, and heatsink. The closed-loop control system includes also a safety mechanism in case of chiller failure. Using this control system, the detection module can reach the optimum temperature of −40°C.Both units (embedding STM32 microcontrollers) provide USB communication to the control PC of the beamline in an integrated user-friendly graphical interface, and are powered by a single +24 V (5 A max.) compact AC-DC module.

The spectrometer is mounted on a motorized translation stage so that the distance between the detector and the target can be adjusted. The connection between the spectrometer and the scattering chamber is realized by an O-ring capable of preserving the vacuum of the scattering chamber. The complete ARDESIA spectrometer solution including its auxiliary electronics is shown in figure 5.

3 Results and discussion

3.1 Measurements at the LNF DAΦNE-light DXR1 beamline

The first measurements with the ARDESIA spectrometer were carried out in the LNF DAΦNE-Light DXR1 soft X-ray beamline [8], providing photons in the energy range of 0.9–3.0 keV. XRF measurements of test samples at RT, such as graphite sheet and Pyrex^{TM} glass, were performed using 4-channel DPP (Digital Pulse Processor) XIA DXP-XMAP with an optimum shaping time of t_{peak} = 2 µs and t_{flat} = 0.1 µs, exploiting the best energy resolution of the instrument. As shown in figure 6 (left), the instrument was able to detect the fluorescence emission of low Z elements down to Carbon (Z = 6, K_{α_1} line = 277 eV). This is a good indication not only of the low-energy response of the silicon detector but also of the low noise and insensitivity of EM interferences of the whole instrument in the synchrotron environment.

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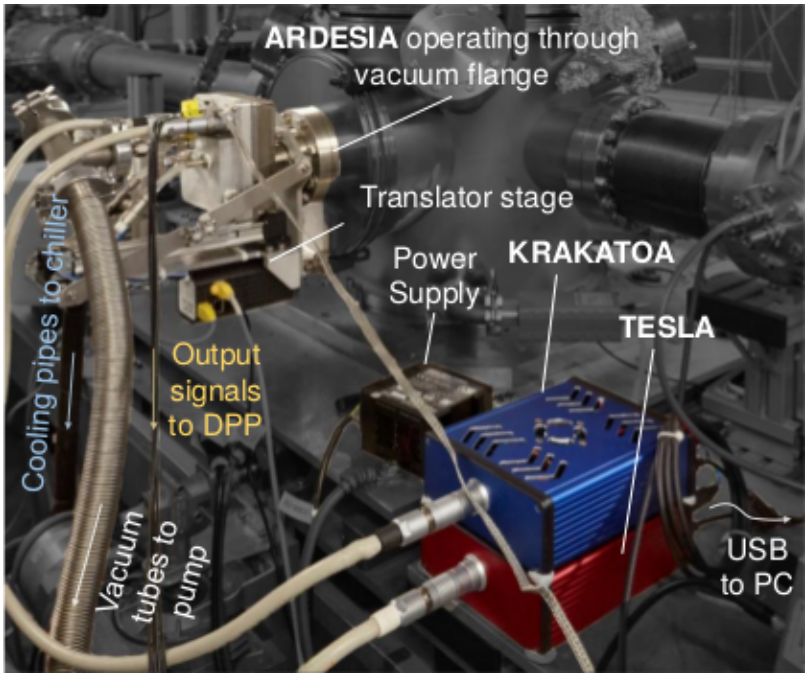


Figure 5. Complete ARDESIA spectrometer system installed on LISA BM-08 beamline.

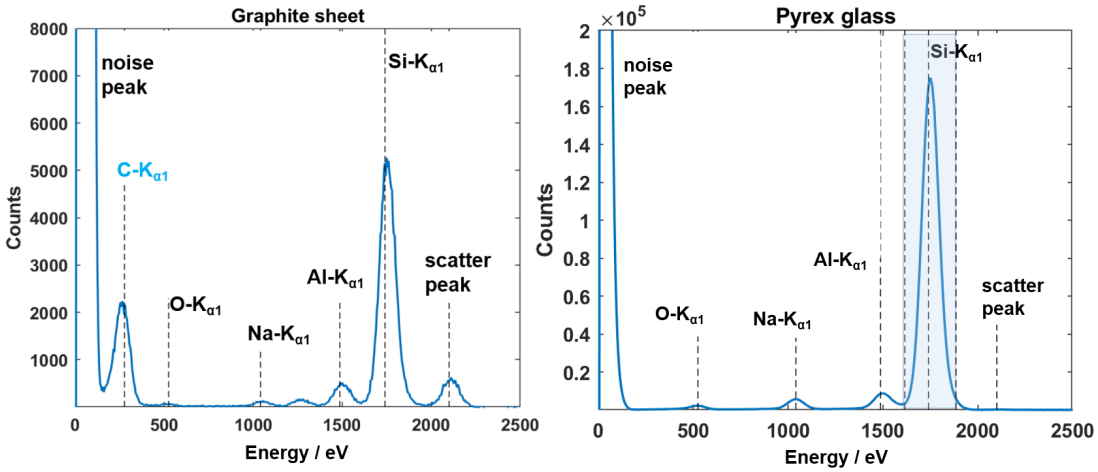


Figure 6. XRF measurement results on graphite sheet (left) and Pyrex^{TM} glass (right), with excitation energy = 2100 eV performed at DXR1 beamline.

XAFS measurements on Pyrex^{TM} glass at RT were performed at the Si K-edge and it was also possible to compare the spectra acquired in transmission mode and in fluorescence mode using ARDESIA. The ROI applied to the Si K_α peak is 1.62–1.89 keV, also described in figure 6 (right). figure 7 shows that the absorption spectra acquired by the 4 different channels are very similar, and that there are no relevant differences between results achieved using the two different modes, confirming the effective operation of the instrument.

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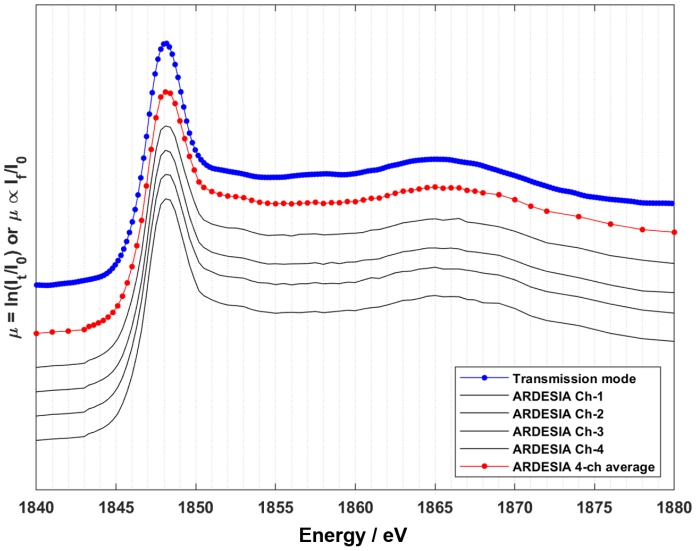


Figure 7. Si-K absorption spectrum from XAFS on Pyrex^{TM} glass using transmission mode and ARDESIA fluorescence mode, performed at DXR1 beamline.

3.2 Measurements at the BM-08 "LISA" CRG beamline at ESRF

Other measurements with ARDESIA instrument were performed at the BM-08 "LISA" CRG beamline at ESRF [19], able to provide high intensity synchrotron radiation within the energy range of 4.9–72 keV. The target of the measurements was to assess the performance of the detector in terms of high throughput capability and stability of the measurements over time. Right after the instrument installation, as a preliminary assessment, XRF measurements of Mn-K$_\alpha$ lines emitted from ^{55}Fe laboratory source were performed and repeated multiple times using 1-channel DANTE-4551 DPP. The measurements were done inside the sample chamber saturated with nitrogen gas. Figure 8 (left) shows that the resolution at the shortest peaking time (t_{peak} = 32 ns, t_{flat} = 128 ns) is 205.2 eV, while at t_{peak} = 3 µs, the optimum energy resolution obtained was 127.3 eV. The results were similar to the ones achieved in the characterization of the detection module used in the instrument which were the one reported in [18]. Thus, it confirms the instrument's immunity against external disturbances, maintaining satisfactory energy resolution even in the synchrotron "noisy" environment.

Figure 8 (right) shows the stability metrics obtained from 60 repeated measurements. Each measurement used t_{peak} = 512 ns and t_{flat} = 128 ns, and it lasted for t_{acq} = 30 s. This test shows that the energy resolution, the centroid of the acquired Mn-K$_\alpha$ spectra, and its OCR remains stable after 60 measurements in about 2 hours, indicated by the small variation ($\sigma < 0.4\%$) of all the measured metrics. These results showing almost constant total counts over ROI, confirm the stability of the instrument, especially for XAFS measurements.

After passing the preliminary assessment, XAFS measurements with synchrotron radiation at RT were performed on two different samples: kesterite and protochabournéite. XAFS measurements in fluorescence mode on a kesterite ($Cu_2(Zn,Fe)SnS_4$) sample were performed at the Cu K-edge, selecting the Cu K$_\alpha$ peak with the ROI 7.7–8.3 keV, shown in figure 9 (left). The measurements were sequentially repeated 4 times, with a total acquisition time of 8 h. The XIA DXP-2X DPP was used with a t_{peak} = 125 ns, t_{flat} = 150 ns, ICR ≈ 50 kcps, and acquisition time t_{acq} = 10 s per point.

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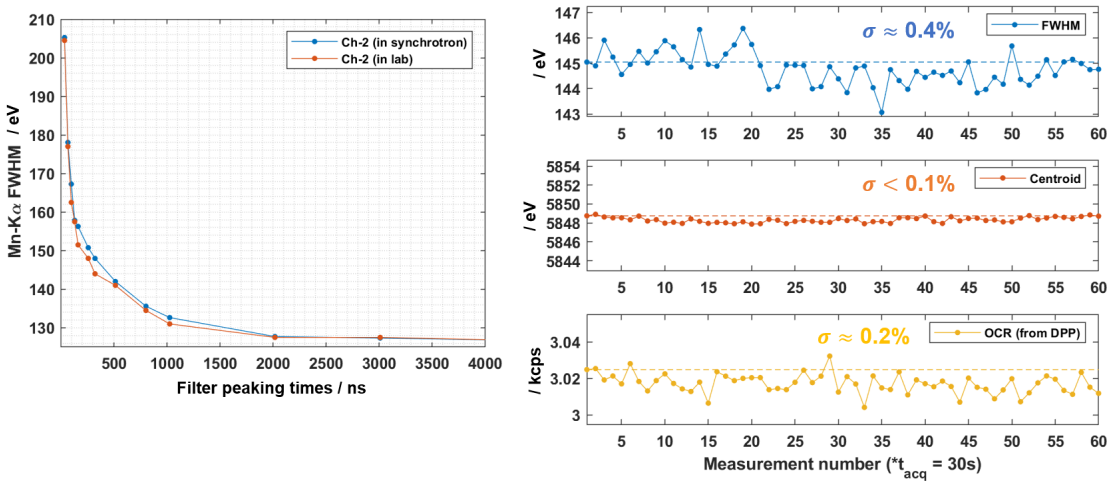


Figure 8. Preliminary measurement results using an ^{55}Fe source: energy resolution (left), and stability metrics (right), performed inside the sample chamber at LISA BM-08 beamline.

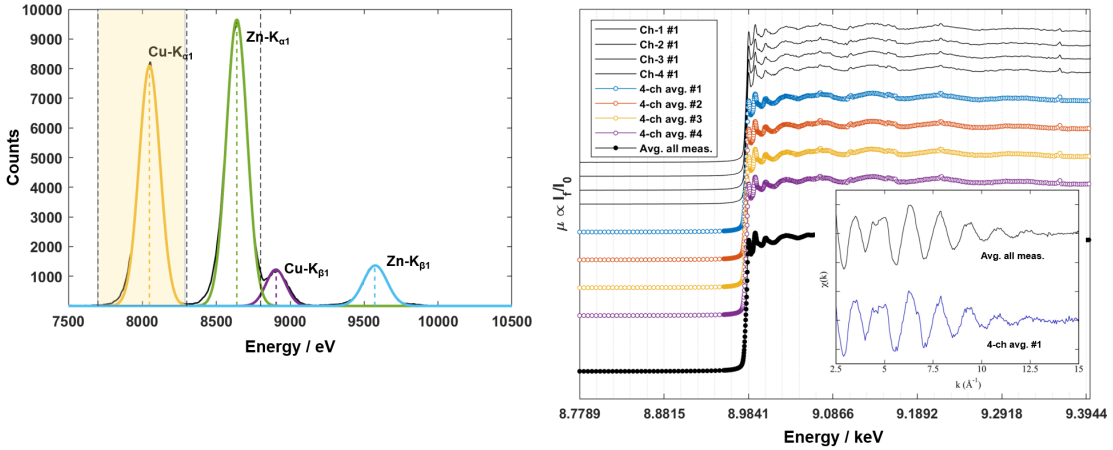


Figure 9. Fluorescence emissions from kesterite (left), Cu K-edge absorption spectra (right), and EXAFS region transformed in k-space (inset), performed at LISA BM-08 beamline.

As shown in figure 9 (right), no significant differences were observed between multiple acquired spectra, further validating the stability of the detector. Repeating several times the measurements allowed to verify reproducibility of data taking and reduce standard deviations of the EXAFS data. From the reconstructed EXAFS spectra (inset of figure 9), it can be seen that the single acquisition obtained summing the 4 channels provide already good results, if compared to the sum of all four 4 channels acquisitions.

XAFS on protochabournéite was performed to demonstrate the capabilities deriving from the very good energy resolution and stability of the instrument. Protochabournéite [20] ($Tl_2Pb(Sb_{9-8}As_{1-2})_{\Sigma 10}S_{17}$) is a complex mineral bearing both Tl and As. Given the proximity of Tl-L_α and As As-K_α emission lines (10.269 and 10.543 keV, respectively), performing measurements at Tl-L_{III} edge is, as shown in figure 10 (top-left), rather complicated and it previously

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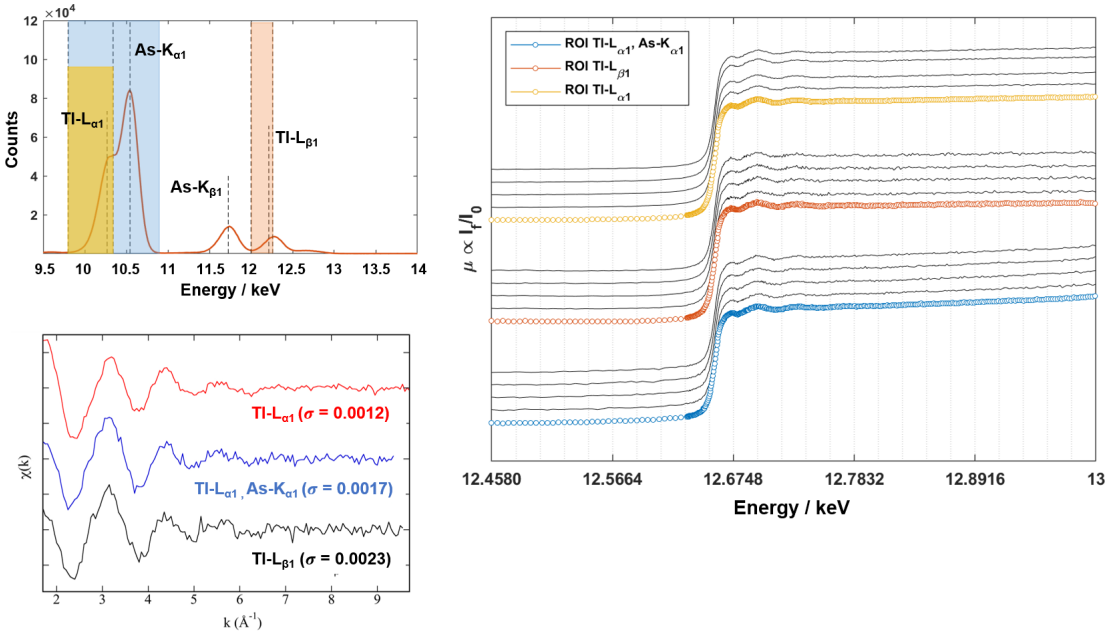


Figure 10. Fluorescence emissions from protochabournéite (top-left), its Tl L_{III}-edge absorption spectra (right), and EXAFS region transformed in k-space (bottom-left), performed at LISA BM-08 beamline.

implicated the use of complex experimental setups for analogous measurements carried out on BM-08 [21]. The good energy resolution of the detector allowed to identify the shoulder of the Tl-L$_\alpha$ line on the side of the more intense As-K$_\alpha$ peak; spectra at the Tl-L_{III} edge were thus measured using three different ROIs: Tl-L$_\alpha$ only (9.80–10.34 keV, yellow), Tl-L$_\alpha$ and As-K$_\alpha$ lines (9.80–10.90 keV, blue), and Tl-L$_\beta$ lines (12.00–12.26 keV, orange), as shown in figure 10 (left). In this measurement, the same DPP was used with t_{peak} = 125 ns, t_{flat} = 150 ns, ICR ≈ 5 kcps, t_{acq} = 10 s per point, and total acquisition time of 1.3 h. The resulted absorption spectra from different ROIs (colored curves), including the spectra obtained from each channels (black curves), are shown in figure 10 (right). The computed EXAFS spectra according to the mentioned ROIs are reported (inset of figure 10), together with the noise level of the EXAFS spectrum quoted by the σ value. Indicated by lower noise of the EXAFS spectrum background, the spectrum formed by Tl-L$_\alpha$ ROI has better quality than the other two spectra, confirming the better choice of this ROI selection.

4 Conclusions

After reporting very preliminary results [22], ARDESIA — a 4-channel SDD-based X-ray detector for high-count rate experiments — has been extensively characterized and the main results have been here reported. The results of the tests and measurements carried out using synchrotron radiation successfully show the full functionality of the instrument, ease of installation and confirm the performance in terms of good energy resolution, high count-rate capability, and stability. Two prototypes of the instrument have been installed at the ESRF LISA BM-08 and at the LNF DAΦNE-Light DXR1 beamline, being available to the users. The spectrometer has demonstrated state-of-the-art performances in terms of achieved energy resolution and counting rates in the operative

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environment of synchrotron beamlines, also on samples of mineralogical interest such as kesterite and protochabournéite. With respect to other detector solutions, we think that ARDESIA has a high potential for increasing relatively easily in number of channels (16 channels in the prototype under development), thanks to the instrinsic modularity of the SDD detector, still maintaining compactness of the detection head. In addition, development of custom detector configurations for synchrotron beamlines can be also foreseen thanks to the versatility of the ARDESIA components.

Further developments include also the production of SDDs with thicker substrates (up to 1 mm) and higher pixel density (from the current 2×2 array with 5 mm pitch to a 4×4 array with both 5 mm and 2 mm pitches) in order to improve, respectively, the detection efficiency at higher X-ray energies and increase the total throughput of the instrument. Interdependently, the development of a multichannel ASIC utilizing very fast analog pulse processing techniques, based on Semi-Gaussian shaping filters with 200 ns filter pulse width, novel peak detection and rejection method is currently ongoing [23]. The ASIC offers the possibility of integrated readout solution inside the instrument to further increase the immunity with respect to external disturbances and reduce the cost of the readout electronics in view of future upscaling of the detection systems to several tens of units, with respect to the adoption of DPP solutions.

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